



Research Paper

Design and multi-objective optimization of efficient UAV battery thermal management system using PCM–air synergistic cooling strategy

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ABSTRACT

Efficient thermal management is critical for the safety and performance of lithium-ion batteries (LIBs) in unmanned aerial vehicles (UAVs). Various battery thermal management systems (BTMS), such as air, liquid, phase change material (PCM), and hybrid cooling, have been developed to address this need. However, their application in UAVs poses a significant challenge, as most systems fail to adequately balance thermal performance with the stringent constraints on weight and energy consumption. Furthermore, a systematic framework for evaluating and optimizing these trade-offs remains underdeveloped in the current literature. To address this, we propose a hybrid BTMS that combines PCM with air cooling. This design integrates the reliability of air cooling with the high latent heat storage capacity of PCM. Single-factor analysis shows that higher PCM thermal conductivity and latent heat improve cooling performance. Moderate increases in battery spacing (≤ 3 mm) and airflow velocity (≤ 4 m/s) also improve cooling performance, however, further increases yield diminishing returns. Furthermore, a multi-factor, multi-level numerical investigation was conducted, considering performance grade of composite PCM, inter-cell spacing, and airflow velocity. Subsequently, a hybrid entropy–subjective weighted technique for order preference by similarity to an ideal solution (TOPSIS) was applied to determine the optimal configuration. The optimized BTMS achieves a 43.3% reduction in the maximum temperature, maintaining it below 46.8 °C, while incurring a modest total mass increase of only 22.2 wt%. This study thus delivers a viable thermal management strategy for UAV applications and, more broadly, proposes a systematic and transferable framework for the design and optimization of next-generation BTMS.

1. Introduction

In recent years, unmanned aerial vehicles (UAVs) have found extensive applications across diverse fields, including aerial photography, logistics and transportation, environmental surveillance, and military reconnaissance [1,2]. As UAV functions expand and performance demands intensify, requirements for onboard energy storage systems have become increasingly stringent. Among various energy storage technologies, the lithium-ion batteries (LIBs) have become the most promising power source for UAVs due to their high energy density, long cycle life, and environmental friendliness. However, during UAVs operation, especially during high-power discharge processes, LIBs generate substantial heat as a byproduct of electrochemical reactions

[3]. Moreover, the optimal working temperature range for LIBs is typically 20 °C to 35 °C, with a maximum allowable temperature difference of no more than 5 °C across the battery pack. Exceeding this temperature range can lead to capacity degradation, shortened lifespan, and in severe cases, thermal runaway, thereby posing safety risks to the UAV system. Therefore, it is essential to develop an effective thermal management strategy to maintain the battery temperature within the safe operating range, ensuring stable performance and flight reliability of UAVs [4,5].

Based on the cooling medium employed, battery thermal management systems (BTMS) can be generally categorized as air cooling [6], liquid cooling [7], phase change material (PCM)-based cooling [8], and hybrid cooling systems. For UAVs, stringent constraints on weight and volume pose challenges to conventional cooling approaches. Air cooling is lightweight and easy to integrate into UAV platforms. However, it fails

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Nomenclature

List of symbols

k	Thermal conductivity of PCM, W/(m·K)
H	Latent heat of PCM, kJ/kg
s	Inter-cell spacing, mm
v	Airflow velocity, m/s
Q_{gen}	Heat generation rate of battery, J
Q_r	Reaction heat of battery, J
Q_p	polarization heat, J
Q_j	Joule heat, J
I	Current of battery, A
T	Temperature of battery, °C
$\partial T/\partial t$	Temperature gradient, K/s
Re	Reynolds number
u_1	Velocity of the coolant, m/s
d_e	Equivalent hydraulic diameter, m
k_b	Thermal conductivity of battery, W/(m·K)
$c_{p,b}$	Heat capacity of battery, J/K
$w_j^{(o)}, w_j^{(s)}$	Objective weight and subject weight
x_{ij}	Original decision matrix
r_{ij}	Normalized decision matrix

w_j	Comprehensive weight fusion
v_{ij}	Weighted matrix
S_i^+, S_i^-	Euclidean distances
C_i	Relative closeness
T_{max}	Maximum temperature, °C

Greek symbols

θ	Melted proportion of PCM
μ_1	Dynamic viscosity, Pa·s
ρ_b	Density of battery, kg/m ³
α	Fusion coefficient
ΔT	Temperature difference, °C

Abbreviations

UAVs	Unmanned aerial vehicles
LIBs	Lithium-ion batteries
PCM	Phase change material
BTMS	Battery thermal management system
TOPSIS	Technique for Order Preference by Similarity to Ideal Solution
PCC	Phase change composite

to provide sufficient cooling under high heat flux, particularly during high-power discharge process. Moreover, liquid cooling provides excellent heat dissipation due to its high heat capacity and thermal conductivity [9,10]. Experimental investigations have demonstrated that liquid immersion cooling can effectively improve temperature uniformity and safety under high heat flux conditions, particularly for stationary and commercial energy storage systems [11–13]. However, such systems generally require dielectric fluids, sealed enclosures, and additional structural components, which increase system mass and complexity. This makes liquid cooling unsuitable for lightweight UAV platforms. PCM-based cooling has attracted growing interest for its capacity to absorb large amounts of latent heat, which enables effective suppression of rapid temperature rises during peak loads. However, the heat-dissipation capacity of PCM is relatively limited, and its thermal storage capability cannot be rapidly recovered once the phase transition is complete [14–17].

To overcome these issues, a hybrid BTMS that combines multiple cooling strategies has been proposed. For example, Pu et al. [18] developed a lightweight hybrid BTMS combining PCM and liquid cooling in a honeycomb structure, and found that hexagonal PCM design improves overall performance while reducing system mass. Zhou et al. [19] designed a PCM-liquid hybrid BTMS with honeycomb cold plates and showed that increasing plate number enhances cooling while maintaining a compact structure and achieving a temperature difference below 2 °C. Chen et al. [20] proposed a wave-shaped cold plate coupled with tubular PCM for cylindrical cells, demonstrating that cross-flow and optimized PCM thickness significantly reduce maximum temperature and improve uniformity. Liu et al. [21] built a hybrid mini-channel cold plate/PCM BTMS, showing that the system outperforms single PCM or liquid cooling while consuming much less energy. Tang et al. [22] developed a coupling system with cooling plates, U-shaped heat pipes, and PCM for 21,700 cells, achieving lower temperature and better uniformity than conventional hybrid or single cooling systems. Khan et al. [23] compared natural air, liquid, PCM, and hybrid cooling for a 5000 mAh pack, and found hybrid cooling yields the lowest maximum temperature and highest safety. Chavan et al. [24] examined different coolants and flow modes, revealing that silicon oil offers superior heat transfer and lower pressure drop than water, glycol, Novec 7000, or mineral oil. Sadeh et al. [25] investigated a direct/indirect hybrid liquid

cooling system for 21,700 batteries and demonstrated that combining immersion and channel cooling with counterflow maintains temperature below 32 °C with high uniformity. Overall, hybrid BTMS have shown promising cooling performance for LIBs, however, most existing designs target large or ground-based platforms with relatively relaxed constraints. There remains a notable lack of lightweight, low-power hybrid thermal management solutions specifically developed for UAV battery systems.

Furthermore, optimization [26–28] of the BTMS is crucial to improving the thermal control efficiency of UAV battery modules. Researchers have devoted extensive efforts to identifying the influential parameters of BTMS to achieve optimal temperature regulation performance through a variety of analytical and optimization methods. Both experimental investigations [29,30] and numerical simulations [31–33] have been employed to examine the temperature field distribution and transient thermal behavior of the battery pack under different operating conditions. For instance, Cheng et al. [34] developed a PCM-water hybrid BTMS and applied deep reinforcement learning for optimization. Compared with NSGA-II and MOPSO, the DRL-based design achieved lower maximum temperature and better temperature uniformity. Su et al. [35] investigated a U-shaped liquid cooling BTMS using CFD and genetic programming. An AI-based NSGA-II optimization was adopted, showing improved temperature control and a significant reduction in pressure drop. Cheng et al. [36] designed a finned heat sink with heat pipes for electronic chassis cooling. Response surface optimization indicated reduced junction temperature (~10 K), enhanced temperature uniformity, and lower thermal resistance. Ou et al. [37] proposed a multi-objective optimization framework for heat sinks in high-power electronics using NSGA-II and entropy-weighted TOPSIS. The optimized design reduced thermal resistance by 15.07% and system volume while being validated experimentally. Sui et al. [38] introduced a hybrid manifold cold plate BTMS for 280 Ah LiFePO₄ batteries. Response surface and NSGA-II optimization maintained battery temperature within safe limits under 3C discharge and outperformed air and immersion cooling. Kang et al. [39] optimized a coupled PCM-liquid cooling BTMS by investigating the effects of cooling channel configuration, PCM thickness, coolant flow rate, and ambient temperature. At 5C discharge, the system achieved $T_{max}=36.13$ °C and $\Delta T = 4.04$ °C, demonstrating improved thermal uniformity compared with single

cooling strategies. Xiang et al. [40] applied a pseudo-3D topology optimization to a hybrid forced PCM cooling BTMS, optimizing cold plate channels considering conjugate heat transfer effects. The optimized design improved heat dissipation while reducing flow power consumption compared with conventional cold plates. Lin et al. [41] performed a multi-objective topology optimization of a liquid-cooled plate for 280 Ah prismatic batteries. The optimized channel design improved heat transfer and thermal uniformity while limiting pressure drop compared with conventional cold plates. Although individual parameters influence thermal behavior and have been widely studied, a comprehensive understanding of the coupled effects of multiple factors is lacking. This coupled understanding is essential for BTMS optimization in UAVs. Moreover, existing multi-parameter analyses predominantly target large-scale systems, with insufficient attention paid to the cylindrical battery packs used in UAVs, where stringent weight and energy constraints are critical.

In this study, an efficient BTMS for cylindrical LIBs in UAV applications is proposed, integrating phase change composite (PCC) with air cooling. The system is designed to mitigate rapid temperature rise during high-rate discharge, while satisfying the stringent constraints on weight and energy consumption inherent to UAV platforms. A coupled transient thermal model of the PCM-air hybrid cooling architecture is developed to characterize its thermal response under high-power operating conditions. Single-factor sensitivity analyses are first conducted to evaluate the influence of thermal conductivity and latent heat of PCC, battery spacing, and airflow velocity on the maximum temperature and temperature uniformity of the battery pack. Subsequently, a multi-factor, multi-level numerical investigation is implemented to investigate the combined effects of performance grade of PCC, inter-cell spacing, and inlet airflow velocity on the thermal performance of the BTMS. To identify the optimal system configuration, an integrated optimization framework combining the entropy weighting method with TOPSIS is adopted to systematically balance the trade-offs among maximum temperature, temperature uniformity, system mass, and fan power consumption under a defined safety constraint. Unlike previous studies that primarily focus on material enhancement or large-scale battery systems, the present work addresses the unique thermal and weight constraints of UAV platforms. The novelty of this study lies in integrating a lightweight PCM-air hybrid cooling architecture with a systematic entropy-subjective weighted multi-objective optimization framework, enabling a balanced trade-off among thermal performance, mass increase, and energy consumption.

2. Model description

2.1. Numerical model

Fig. 1 exhibits the thermal-management configuration of the UAV battery system. As shown in Fig. 1(a), the UAV carries several battery modules composed of cylindrical LIBs. To reduce complexity, this study selects a battery pack as research subject, which consists of nine LIBs arranged in a 3×3 configuration. Besides, each cylindrical LIB is simplified as a smooth cylinder without explicitly modeling the battery terminals or minor structural details. This assumption is commonly adopted in battery thermal management simulations when the focus is on module-level thermal behavior [42–44]. Since the present study aims to compare the relative thermal performance of different BTMS configurations under consistent conditions, such geometric simplification does not affect the overall temperature distribution trends or the validity of the parametric analysis. As demonstrated in Fig. 1(b, c), the inter-cell gaps are filled with PCC, which absorb heat during high-rate discharge. To prevent complete melting of the PCC and the consequent decline in cooling capability, an aluminum U-shaped air-cooling plate is employed to dissipate the stored heat in a timely manner, as illustrated in Fig. 1. Notably, the PCM considered in this study represents a form-stable PCC with structural stabilization provided by polymer matrices and conductive fillers. In such systems, molten PCM mobility is significantly restricted, and heat transfer is predominantly governed by conduction [45,46]. Therefore, leakage and internal natural convection effects were neglected in the present system-level modeling framework [47–49].

Since the PCC is filled in the inter-cell regions, the spacing directly determines the amount of material that can be incorporated. Based on Zhang et al. [50], who adopted a 1 mm gap between cylindrical batteries, the initial spacing in this work is set to 1 mm to ensure sufficient PCC filling while maintaining high pack energy density. The air-cooling plate has a sidewall thickness of 0.5 mm, a channel thickness of 2 mm, and a total structural thickness of 3 mm. Air is selected as the coolant, as UAVs inherently generate strong airflow during flight, enabling passive heat removal and minimizing additional energy consumption. In the geometric model, the LIBs are simplified as smooth cylinders without terminals or other minor features. In this hybrid cooling design, both PCC and air serve as heat dissipation media. Their thermophysical properties significantly influence the thermal and structural behavior of the UAV battery pack. Four key parameters, including thermal conductivity (k) and latent heat (H) of PCC, inter-cell spacing (s), and airflow velocity (v), are identified as dominant factors governing the maximum temperature and temperature uniformity of LIBs, total system

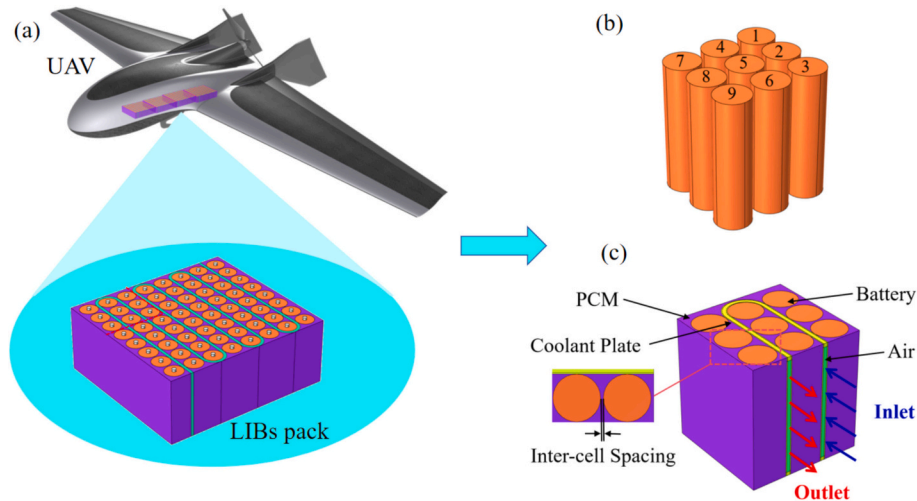


Fig. 1. Overall schematic of the UAV battery hybrid thermal-management system. (a) UAV and its battery thermal management system. (b) Battery pack without cooling. (c) Hybrid thermal management system of LIBs.

weight, and fan power consumption. It should be noted that enhancing the thermal conductivity of phase change materials is often accompanied by a reduction in latent heat, as the introduction of thermally conductive fillers or interfacial modifiers may partially replace the phase change component. Therefore, a trade-off generally exists between thermal conductivity and latent heat capacity. To address this issue, recent studies have focused on material design strategies that enhance heat transfer efficiency while minimizing the sacrifice of latent heat, such as constructing efficient interfacial thermal pathways or employing low-content, selectively distributed conductive fillers [51]. These approaches aim to achieve a balanced improvement in thermal conductivity without significantly degrading the heat storage capability of PCC, which is particularly desirable for lightweight battery thermal management systems. Besides, the selected design variables-including PCC thermal properties, inter-cell spacing, and airflow velocity-represent the most practically adjustable parameters in UAV BTMS applications. The study focuses on establishing a clear and engineering-oriented optimization framework under controlled parameter ranges, rather than expanding the design space to include all geometric and flow configuration variables.

Therefore, this study systematically investigates the coupled effects of these parameters to evaluate the thermal response, temperature distribution, structural characteristics, and overall energy efficiency of the proposed BTMS. Numerical analysis was conducted to provide a comprehensive understanding of thermal behavior of UAV battery pack. The analysis included both single-factor sensitivity studies and multi-factor, multi-level numerical experiments to examine the coupled influence of key design parameters on the thermophysical and energy performance of BTMS. All numerical experiments were implemented in COMSOL Multiphysics to evaluate the transient temperature response and parameter interactions under realistic operating conditions. The thermophysical parameters employed in the numerical model are summarized in Table 1 [52–56].

2.2. Mathematical model

2.2.1. Battery heat generation model

The overall heat generation of the battery, denoted as Q_{gen} , consists of three distinct contributions: reaction heat (Q_r), polarization heat (Q_p), and Joule heat (Q_j) [57,58]. These heat sources can be quantitatively described by the corresponding expressions.

$$Q_{\text{gen}} = Q_r + Q_p + Q_j \quad (1)$$

$$Q_r = -T\Delta S \frac{I}{nF} \quad (2)$$

$$Q_p = I^2(R_t - R_e) = I^2R_p \quad (3)$$

Table 1
Physical parameters of PCC and batteries [52–56].

Specifications	Value
Type of battery	Panasonic NCR18650GA
Cut-off voltage of battery	3.6–4.2 (V)
Nominal capacity of battery	3.4 (Ah)
Height of battery	65 (mm)
Thermal conductivity of battery in radial direction	3.4 (W/(m·K))
Thermal conductivity of battery in polar angle	34 (W/(m·K))
Thermal conductivity of battery in axial direction	34 (W/(m·K))
Density of battery	2800 (kg/m ³)
Specific heat of battery	950 (J/(kg·K))
Thermal conductivity of PCC, solid	0.425 (W/(m·K))
Thermal conductivity of PCC, liquid	0.152 (W/(m·K))
Density of PCC, solid	814 (kg/m ³)
Density of PCC, liquid	771 (kg/m ³)
Melting temperature of PCC	35.69 (°C)
Latent heat of PCC	247.05(kg/kJ)

$$Q_j = I(E-V) = I^2R_e \quad (4)$$

In these formulations, T is the temperature of the battery. ΔS represents the entropy variation of the electrochemical reaction, I is the current, n denotes the number of electrons involved in the reaction, and F is Faraday's constant. Additionally, R_e indicates the ohmic resistance, R_p refers to the polarization resistance, R_t is the total internal resistance. Besides, E corresponds to the open-circuit voltage, and V is the terminal operating voltage.

2.2.2. Heat transfer model

The transient heat transfer behavior within the batteries can be described by the following governing equation:

$$\rho_b c_{p,b} \frac{\partial T}{\partial t} = \nabla \cdot (k_b \nabla T) + Q_{\text{gen}} \quad (5)$$

where ρ_b , $c_{p,b}$, and k_b represent the density, specific heat capacity, and thermal conductivity of the battery material, respectively. Q_{gen} denotes the volumetric heat generation rate inside the battery.

The heat transfer of phase change material is determined by the following equations:

$$\rho_{\text{PCM}} \frac{\delta H_{\text{PCM}}}{\delta t} = \nabla \cdot (k_{\text{PCM}} \nabla T_{\text{PCM}}) \quad (6)$$

$$H_{\text{PCM}} = \int_{T_0}^{T_{\text{PCM}}} c_{\text{PCM}} dT + \theta L \quad (7)$$

$$\theta = \begin{cases} 0, T_{\text{PCM}} < T_s(\text{solid}) \\ \frac{T_{\text{PCM}} - T_s}{T_L - T_s}, T_s \leq T_{\text{PCM}} < T_L(\text{solid/liquid}) \\ 1, T_{\text{PCM}} \geq T_L(\text{liquid}) \end{cases} \quad (8)$$

where ρ_{PCM} , H_{PCM} , and k_{PCM} refer to the material density, enthalpy, and thermal conductivity of PCM. T_{PCM} denotes the temperature of the PCM, while T_0 stands for the ambient temperature. The term θ represents the proportion of PCC that has melted. T_s and T_L correspond to the onset and completion temperatures of the phase transition, which indicate the solid and liquid states, respectively.

As for the convective heat transfer inside the channel, the Reynolds number (Re) is calculated as:

$$Re = \frac{d_e u_1 \rho_1}{\mu_1} \quad (9)$$

where u_1 denotes the velocity of the coolant, ρ_1 is its density, and μ_1 represents the dynamic viscosity.

The equivalent hydraulic diameter of the coolant passage, d_e , can be expressed as:

$$d_e = \frac{2l_1 l_2}{l_1 + l_2} \quad (10)$$

where l_1 and l_2 correspond to the length and width of the flow channel section, respectively. In the present work, the maximum Reynolds number reaches 1240 when the coolant velocity is 5 m/s, indicating that the flow regime remains laminar.

The governing equations for the incompressible laminar flow and heat transfer are given as follows:

$$\frac{\partial \rho_w}{\partial t} + \nabla \cdot (\rho_w \mathbf{v}) = 0 \quad (11)$$

$$\frac{\partial (\rho_w \mathbf{v})}{\partial t} + \nabla \cdot (\rho_w \mathbf{v} \mathbf{v}) = -\nabla \rho_w + \mu \nabla^2 \mathbf{v} \quad (12)$$

$$\frac{\partial (\rho_w c_w T_w)}{\partial t} + \nabla \cdot (\rho_w c_w T_w \mathbf{v}) = \nabla \cdot (k_w \nabla T_w) \quad (13)$$

where, ρ_w , v , μ , c_w , T_w , and k_w denote the density, velocity, dynamic viscosity, specific heat capacity, temperature, and thermal conductivity of the coolant, respectively.

2.2.3. Initial and boundary condition

As illustrated in Fig. 1(b, c), the battery pack is composed of nine 18,650 LIBs, labeled from 1 to 9, with the spaces between them filled with PCC. The spacing between adjacent batteries is a variable parameter, with an initial value of 1 mm. The initial overall dimensions of the battery pack are 61 mm in width and 62 mm in length. Both the initial temperature of the pack and the inlet air temperature are set to 298.15 K. The thermal conductivity and latent heat of the PCC is initially defined as 3 W/(m·K) and 200 kJ/kg. The inlet airflow velocity is 5 m/s. The external surfaces of the battery pack are assumed to be adiabatic. Heat generation inside the batteries acts as the internal heat source. During operation, the heat produced by the batteries is first absorbed by the PCC, then transferred to the cooling plate, and finally removed by the flowing air.

3. Evaluation method

3.1. Multi-factor and multi-level experimental design

Multi-factor and multi-level experiments were designed to analyze the effects of multiple parameters on the BTMS. This approach allows the combined influence of several structural and thermal parameters to be evaluated efficiently without relying on a full factorial design. Three key parameters, including performance grade of PCC, inter-cell spacing (s), and airflow velocity (v), are selected as the main influencing factors. Each parameter is assigned several levels according to its practical range in UAV applications. A total of 60 cases covering different combinations of these parameters are conducted, to investigate their effects on the temperature distribution, system mass, and energy consumption of the hybrid BTMS.

3.2. Entropy-subjective weighted TOPSIS model

Common multi-objective decision-making methods such as AHP, VIKOR, and Pareto-based evolutionary algorithms have been widely used in thermal system optimization [38,59]. In this study, the Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) [37,60] was selected due to its computational efficiency, clarity in handling discrete alternatives, and suitability for integrating both objective and subjective weighting factors in practical engineering decision-making. TOPSIS is an effective multi-criteria decision-making method that ranks alternative schemes according to their relative closeness to an ideal solution. In this method, the optimal scheme should have the shortest distance to the positive ideal solution and the farthest distance from the negative ideal solution. However, the traditional TOPSIS model relies on predefined subjective weights, which may introduce bias, while purely objective weighting approaches, such as the entropy weight method, may ignore expert experience and practical importance of indicators. To overcome these limitations, this study adopts a hybrid weighting strategy that combines entropy-based objective weights with subjective weights derived from engineering judgment. This fusion ensures that both the data-driven information content and expert knowledge are considered, leading to a more balanced and reliable evaluation [37,60].

Let $w_j^{(o)}$ and $w_j^{(s)}$ represent the objective weight obtained from the entropy method and the subjective weight assigned by expert evaluation, respectively. The final comprehensive weight of each indicator is calculated using a fusion coefficient as follows:

$$w_j = \alpha w_j^{(o)} + (1 - \alpha) w_j^{(s)} \quad (14)$$

where α reflects the relative importance assigned to the objective and subjective components. When $\alpha = 1$, the weighting is fully objective, while $\alpha = 0$ represents a fully subjective evaluation.

The detailed procedure of the entropy-subjective weighted TOPSIS method is summarized below:

(1) Data normalization

Given the original decision matrix $X = [x_{ij}]_{m \times n}$, normalization is performed to eliminate dimensional differences:

$$r_{ij} = \frac{\max(x_j) - x_{ij}}{\max(x_j) - \min(x_j)} \quad (15)$$

(2) Determination of objective and subjective weights

(3) The entropy method determines the objective weight $w_j^{(o)}$ based on data dispersion:

$$p_{ij} = \frac{r_{ij}}{\sum_{i=1}^m r_{ij}} \quad (16)$$

$$E_j = -k \sum_{i=1}^m p_{ij} \ln(p_{ij}) \quad (17)$$

$$d_j = 1 - E_j \quad (18)$$

$$w_j^{(o)} = \frac{d_j}{\sum_{j=1}^n d_j} \quad (19)$$

Expert judgment or prior experience assigns a subjective weight $w_j^{(s)}$ to each indicator, ensuring

$$\sum w_j^{(s)} = 1 \quad (20)$$

Comprehensive weight fusion:

$$w_j = \alpha w_j^{(o)} + (1 - \alpha) w_j^{(s)} \quad (21)$$

(4) Weighted normalized matrix

Construct the weighted matrix $V = [v_{ij}]$:

$$v_{ij} = w_j \times r_{ij} \quad (22)$$

(5) Determine the ideal and negative-ideal solutions

$$v_j^+ = \max(v_{ij}), v_j^- = \min(v_{ij}) \quad (23)$$

(6) Compute the Euclidean distances

$$S_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^+)^2} \quad (24)$$

$$S_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - v_j^-)^2} \quad (25)$$

(7) Evaluate the relative closeness

$$C_i = \frac{S_i^-}{S_i^+ + S_i^-}, 0 \leq C_i \leq 1 \quad (26)$$

A larger C_i value indicates that the scheme is closer to the ideal solution, representing better overall performance.

(8) Ranking and decision

Finally, all schemes are ranked according to C_i . The optimal parameter combination corresponds to the scheme with the highest relative closeness, reflecting the best compromise between thermal performance, system mass, and energy consumption in the UAV battery pack.

It should be noted that the present study does not aim to develop a novel optimization algorithm. Instead, the entropy-subjective weighted TOPSIS method is employed as an engineering decision-making tool to rank discrete design combinations under multi-objective constraints. Compared with continuous evolutionary algorithms (e.g., NSGA-II), which are more suitable for continuous parameter optimization and Pareto front generation, TOPSIS provides a transparent ranking mechanism that directly yields an implementable optimal configuration in discrete design scenarios.

4. Experimental validation

4.1. Grid independence

A grid independence test was carried out to ensure that numerical results were not influenced by the mesh density of the computational domain. As shown in Fig. 2, several mesh configurations were examined by progressively increasing the total number of elements from 44,011 to 475,575, while monitoring the corresponding maximum temperature of the battery module. The results indicate that the variation in maximum temperature becomes negligible (within 1%) once the mesh size exceeds 191,834 elements. This finding confirms that further mesh refinement has minimal impact on the thermal predictions and that the model has achieved grid convergence. Accordingly, the mesh containing 191,834 elements was selected for all subsequent numerical simulations, providing an appropriate balance between computational accuracy and efficiency.

4.2. Verification for numerical model of battery

To verify the accuracy of the established model, the temperature variation of a single 18,650 LIB during different discharge rates (including 1C, 2C, and 3C) was examined. The computational results were compared with experimental data [52]. Besides, the numerical experiments were conducted under the same geometric, material, and boundary conditions as those described in the literature. The convective heat transfer coefficient between the battery surface and the surrounding air was set as $7 \text{ W}/(\text{m}^2 \cdot \text{K})$ according to natural convection conditions.

As shown in Fig. 3, the simulated temperature profile of the 18,650 battery during different discharge rates exhibits excellent agreement

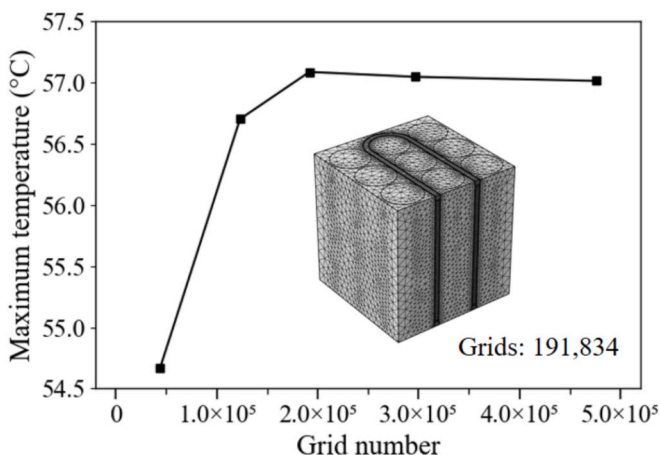


Fig. 2. Grid independence verification and grid model for numerical analysis.

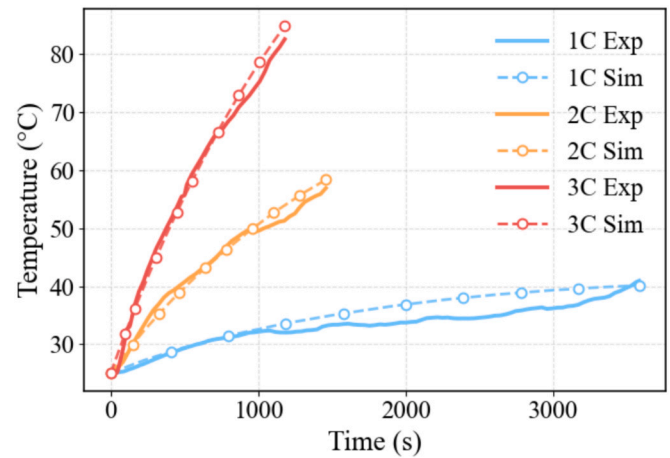


Fig. 3. Verification for heat generation of battery [42].

with the experimental results obtained by Lai et al. [52]. The numerical computation successfully captures both the heating rate and the final temperature plateau. Throughout the 3C discharge period, the temperature difference between numerical analysis and experiment remains small, generally within $\pm 4^\circ \text{C}$. The calculated mean absolute percentage error (MAPE) is 2.78%, confirming the high accuracy of the developed thermal model. In addition, supplementary battery tests were conducted in this work to further evaluate the applicability of the model under practical thermal management conditions. The comparison between the experimental measurements and numerical predictions for a representative 1C discharge and 2C charge cycle is provided in the Supporting Information (Fig. S5). Also, the experimentally measured temperature evolution under continuous cyclic charge/discharge conditions is further presented in Fig. S6. Details of the experimental setup used for the supplementary battery thermal management tests are described in the Supporting Information (Supplementary Note 3 and Fig. S4). The agreement between simulation and experiment in Fig. S5, together with the continuous cycling experimental results shown in Fig. S6, further supports the applicability of the numerical method for predicting transient battery temperature evolution under practical thermal management conditions. Therefore, the model is reliable for subsequent thermal management optimization and parametric studies.

5. Results and discussion

5.1. Single-factor influence analysis of key parameters

Four primary factors were examined: the thermal conductivity (k) and latent heat (H) of the PCC, the spacing between adjacent batteries (s), and the airflow velocity (v). The baseline values for these parameters were set to $3 \text{ W}/(\text{m} \cdot \text{K})$, $200 \text{ kJ}/\text{kg}$, 1 mm , and $5 \text{ m}/\text{s}$, respectively. Fig. 4 illustrates the variations in the maximum temperature and temperature difference of the battery module under different conditions. Specifically, the thermal conductivity of the PCC exhibits a pronounced effect on heat dissipation (Fig. 4a). As the thermal conductivity increases, heat conduction within the PCC becomes more efficient, resulting in a lower peak temperature and improved temperature uniformity across the battery pack. As shown in Fig. 4(b,c), increases in latent heat (H) and inter-cell spacing (s) enhance the heat absorption capacity of the PCC, thereby reducing the maximum temperature of the module. However, when exceeds 3 mm, the rate of temperature reduction diminishes, which suggests that the additional PCC volume no longer contributes effectively to cooling. This is because its performance becomes limited by the extended heat conduction path and the reduced utilization efficiency of the material. Overall, the system's heat transfer capacity is constrained by the internal thermal resistance of the PCC and the cooling capability

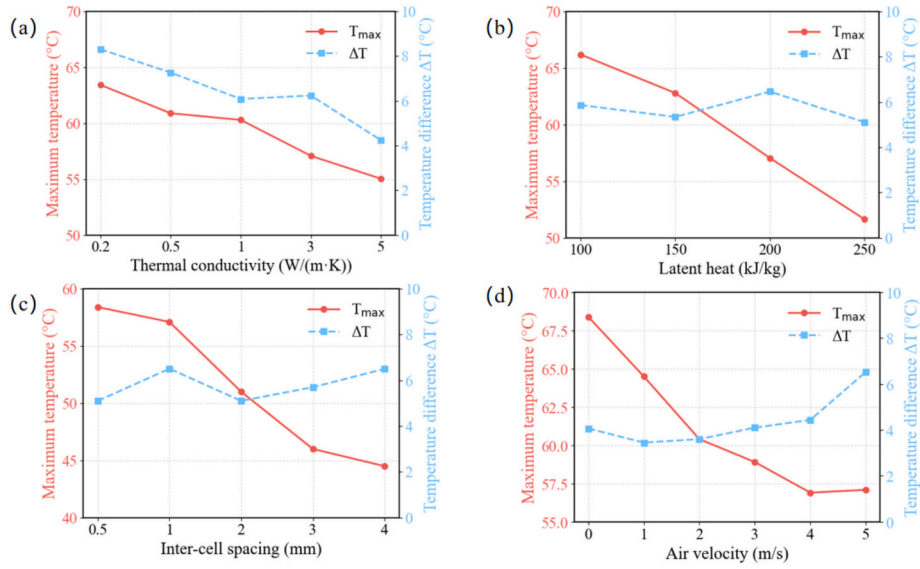


Fig. 4. Effect of factors in different levels on the maximum temperature and temperature difference. (a) Thermal conductivity of PCC. (b) Latent heat of PCC. (c) Inter-cell spacing. (d) Air velocity.

of the airflow. As shown in Fig. 4(d), increasing airflow velocity (v) significantly decreases the maximum temperature (T_{max}), particularly as the velocity rises from 0 to 4 m/s. This trend indicates that stronger airflow enhances convective heat transfer, accelerating the removal of heat accumulated within the PCC and on the battery. However, when the airflow velocity exceeds 4 m/s, the reduction in T_{max} becomes noticeably smaller, suggesting a diminishing return as the system approaches a convective cooling limit. At this point, the thermal resistance on the PCC side becomes the dominant factor governing heat transfer, and further increases in airflow yield only marginal improvements in temperature reduction. In contrast, the temperature difference (ΔT) initially remains nearly constant but begins to increase slightly at higher airflow velocities. This behavior can be attributed to uneven cooling induced by intensified convection near the inlet, which creates localized temperature gradients within the module. Therefore, selecting an appropriate airflow velocity is essential to balance the trade-off between

heat dissipation efficiency and temperature uniformity.

In addition, Fig. 5(a) illustrates the temperature distribution of the battery module at the end of the discharge process, under varying PCC thermal conductivities. As the conductivity increases from 0.2 to 5 $\text{W}/(\text{m}\cdot\text{K})$, the overall module temperature decreases substantially, and the temperature field becomes more uniform. This enhancement occurs because higher thermal conductivity promotes more effective heat transfer from the batteries into the PCC, enabling more uniform heat dissipation. However, once the conductivity exceeds approximately 3 $\text{W}/(\text{m}\cdot\text{K})$, further improvements in temperature distribution become marginal. This indicates that the dominant limitation in heat dissipation has shifted from conduction within the PCC to other mechanisms, particularly convective cooling. Besides, Fig. 5(b) depicts the influence of airflow velocity on the temperature field at the end of the discharge process. At low velocities (1–2 m/s), heat tends to accumulate around the batteries, resulting in relatively high temperatures and noticeable

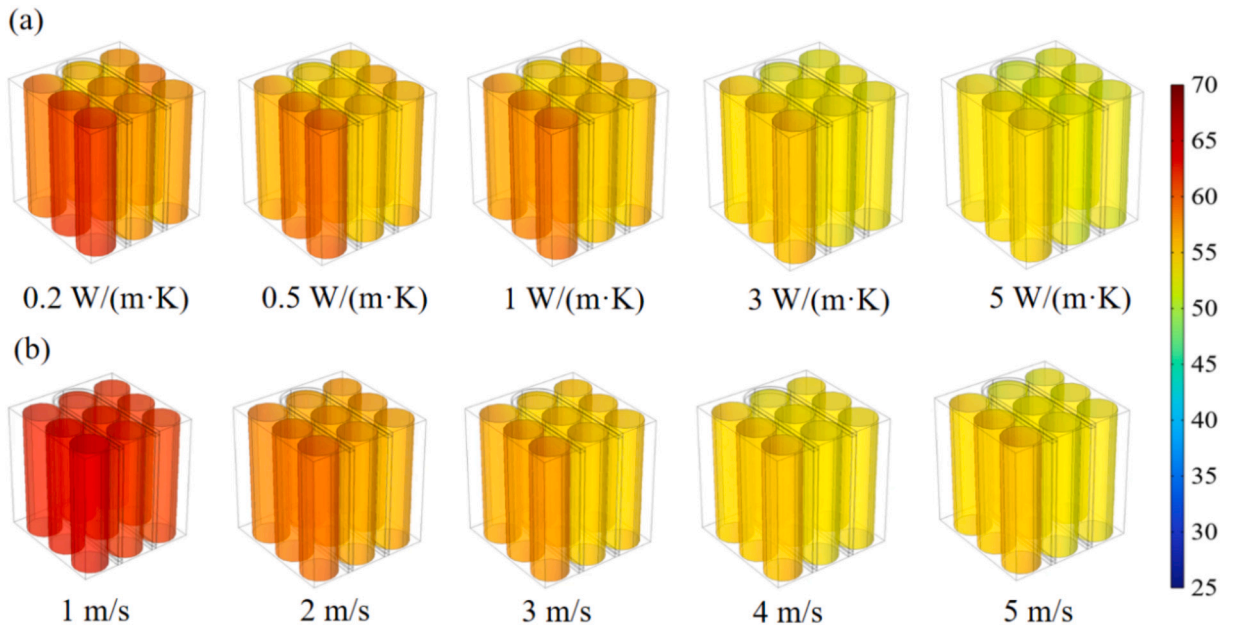


Fig. 5. Temperature distribution of the battery module under different thermal conductivity of PCC and air velocity.

nonuniformity. As the airflow velocity increases to 3–5 m/s, the overall temperature decreases significantly, and high-temperature regions on the battery surfaces gradually diminish. This demonstrates that stronger airflow enhances convective heat transfer and accelerates heat removal from both the PCC and the battery surfaces. Furthermore, the effect of inter-cell spacing is shown in Fig. 6. When the spacing between adjacent batteries increases from 1 mm to 4 mm, the PCC volume fraction increases, allowing more heat to be absorbed during the melting process. Consequently, both the overall temperature and local hotspots decrease markedly. However, as the spacing continues to enlarge, the trend of temperature reduction weakens, suggesting that additional PCC volume no longer contributes effectively to cooling due to the longer heat conduction paths within the PCC.

Fig. 7 shows the evolution of the average temperature of the nine batteries and the entire pack, along with temperature fields at selected time points. The system first undergoes a period of rapid sensible heating (0–200 s), during which the temperature rises from ambient conditions to approximately 30–32 °C, and the temperature curves of all batteries overlap, indicating a highly uniform thermal field. As the PCC begins to melt (200–800 s), latent-heat absorption becomes the dominant mechanism, causing the temperature to increase slowly and nearly linearly. The corresponding temperature distributions remain cool and homogeneous, confirming that the PCC effectively redistributes heat throughout the pack. After roughly 800 s, the latent-heat capacity becomes largely depleted, the temperature rise accelerates, and slight temperature divergence among cells appears. This behavior reflects a shift in the dominant thermal resistance from conduction within the PCC to convection on the air side. Overall, the baseline design maintains excellent temperature uniformity throughout most of the discharge period, with noticeable divergence emerging only as the system approaches thermal saturation. This suggests that the PCC provides effective thermal buffering for roughly 600–800 s, and that extending this duration would require increasing PCC capacity, improving its thermal conductivity, or enhancing external cooling.

5.2. Multi-factor and multi-level experiments

Based on the results of the single-factor sensitivity analysis, a multi-factor and multi-level study was conducted to further evaluate the combined effects of key parameters on the thermal performance of the battery pack. Four different PCC performance grades were selected according to previous studies [61–64]. Notably, representative PCCs were prepared using a flexible polymer-supported paraffin system with thermally conductive additives to provide experimental validation of the

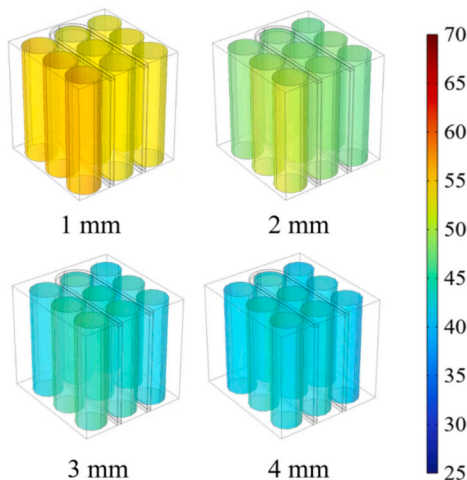


Fig. 6. Temperature distribution of the battery module under different inter-cell spacing.

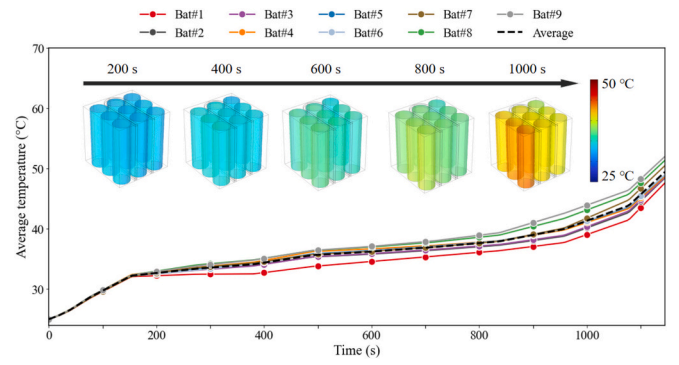


Fig. 7. Time evolution of average temperature for each battery.

material performance. The preparation procedure and compositions of the PCC samples are provided in the Supporting Information (Supplementary Note 1, Fig. S1 and Table S1). Additional thermophysical characterization of the PCC, including phase transition behavior and thermal properties, is presented in the Supporting Information (Supplementary Note 2, Fig. S2 and Table S2). Besides, the thermal cycling stability of the PCC was also evaluated through repeated heating-cooling tests, confirming stable performance during long-term operation (Supplementary Note 4 and Fig. S3). Furthermore, three influencing factors were adopted: the performance grade of PCC, the inter-cell spacing (s), and the air velocity (v). Each factor was assigned multiple levels, including four types of the PCC, three types of the spacing, and five types of the air velocity, resulting in a total of 60 simulation cases. The representative thermophysical property grades of PCC adopted for BTMS optimization in this study are summarized in Table 2. PCC Grade I–IV differ in thermal conductivity, latent heat, and density, covering the typical performance range of PCC reported in the literature. These grades are introduced using a discrete parameter design strategy to systematically evaluate the influence of different conductivity and latent heat levels on the thermal behavior of the BTMS under UAV engineering constraints. Therefore, the values listed in Table 2 should be interpreted as representative performance levels for sensitivity and optimization analysis rather than as exact properties of a specific commercial material. The selected conductivity range (0.2–5 W/(m·K)) reflects the enhancement levels commonly achieved by conductive fillers such as graphite, metal foam, and carbon-based additives, thereby ensuring engineering representativeness.

The values of the three different factors are summarized in Table 3. For each combination, a three-dimensional coupled heat transfer model was established and implemented in COMSOL Multiphysics.

The experimental results included three key thermal performance indicators: the maximum temperature ($T_{\max}$), the maximum temperature difference (ΔT), and the total weight (m) and air velocity (v) of the thermal management system (Table 4). These 60 datasets provide a comprehensive basis for subsequent multi-objective evaluation and optimization using the entropy–subjective weighted TOPSIS method. It should be noted that the present model does not explicitly account for liquid-phase convection within PCM during melting. Future studies may incorporate multiphase flow modeling to further examine detailed melting dynamics.

Table 2

Representative thermophysical property grades of PCC for BTMS optimization.

PCC Grade	Thermal Conductivity (W/(m·K))	Latent heat (kJ/kg)	Density (g/cm ³)	Ref.
Grade I	0.2	250	0.8	[61]
Grade II	1	200	0.85	[61–64]
Grade III	3	150	0.85	[64]
Grade IV	5	125	0.95	[64]

Table 3

Values of different factors.

	Spacing(mm)	PCC Grade	Velocity(m/s)
1	1	Grade I	1
2	2	Grade II	2
3	3	Grade III	3
4	\	Grade IV	4
5	\	\	5

5.3. Comprehensive evaluation based on TOPSIS model

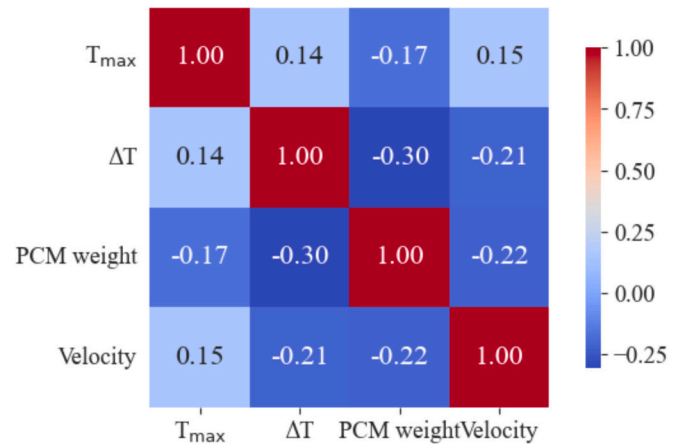
5.3.1. Data analysis

Before performing the multi-objective evaluation, a correlation analysis was carried out to examine the interrelationships among the selected performance indicators. Four evaluation metrics were considered in this study: maximum temperature ($T_{\max}$), temperature difference (ΔT), PCM weight, and air velocity. To assess their dependencies and potential redundancy, the Pearson correlation coefficients were calculated based on the results of the 60 cases, and the outcomes were visualized as a correlation heatmap, as shown in Fig. 8. As illustrated in the figure, $T_{\max}$ shows a weak positive correlation with both ΔT and velocity, while PCM weight exhibits a slight negative correlation with $T_{\max}$. The moderate negative correlation between ΔT and PCM weight ($r = -0.30$) indicates that lighter systems tend to exhibit larger temperature variations, reflecting a trade-off between thermal uniformity and system mass. Overall, the correlation coefficients are relatively small ($|r| < 0.3$), suggesting that the selected indicators are largely independent and can be jointly used in subsequent multi-objective optimization without significant redundancy. It is important to clarify that the Pearson correlation matrix in Fig. 8 does not describe the relationship between design variables and performance objectives, but rather the interdependence among evaluation indicators. The influence of design variables on thermal performance has been comprehensively demonstrated through the single-factor and multi-factor analyses presented in Sections 5.1 and 5.2, where noticeable variations in maximum temperature ($T_{\max}$),

temperature difference (ΔT) were observed across the tested parameter ranges. Therefore, the weak correlations in Fig. 8 do not imply weak physical influence of the design variables, but instead confirm the independence of evaluation metrics used in the multi-objective decision-making process.

5.3.2. Entropy-subjective weighted fusion

To achieve a more comprehensive and balanced weighting scheme, both objective and subjective information were integrated through an entropy-subjective weighting approach. First, the entropy method was applied to calculate the entropy values and objective weights of the four evaluation indicators: maximum temperature ($T_{\max}$), PCM weight, temperature difference (ΔT), and air velocity. The subjective weights were determined based on engineering practice and design priorities in UAV battery thermal management, where thermal safety and light-weight design are generally regarded as the most critical factors. Spe-

**Fig. 8.** Correlation heatmap of four evaluation metrics.**Table 4**

Results of multi-factor and multi-level experiments.

PCC Grade	Spacing (mm)	Velocity (m/s)	$T_{\max}$ (°C)	ΔT (°C)	PCM weight (g)	PCC Grade	Spacing (mm)	Velocity (m/s)	$T_{\max}$ (°C)	ΔT (°C)	PCM weight (g)
Grade I	1.0	1.0	62.9	4.9	45.9	Grade II	1.0	1.0	68.3	3.6	48.8
	1.0	2.0	59.7	5.6	45.9		1.0	2.0	63.5	3.9	48.8
	1.0	3.0	62.6	7.7	45.9		1.0	3.0	65.2	4.6	48.8
	1.0	4.0	58.0	7.4	45.9		1.0	4.0	63.1	4.7	48.8
	1.0	5.0	56.9	7.3	45.9		1.0	5.0	61.7	5.2	48.8
	2.0	1.0	57.5	5.6	64.4		2.0	1.0	63.8	4.3	68.5
	2.0	2.0	58.7	10.8	64.4		2.0	2.0	57.1	4.4	68.5
	2.0	3.0	49.3	6.2	64.4		2.0	3.0	58.5	3.9	68.5
	2.0	4.0	47.9	5.3	64.4		2.0	4.0	55.9	4.3	68.5
	2.0	5.0	48.1	6.5	64.4		2.0	5.0	49.5	10.3	68.5
	3.0	1.0	50.5	5.5	83.8		3.0	1.0	54.7	4.9	89.0
	3.0	2.0	47.9	4.2	83.8		3.0	2.0	49.9	7.7	89.0
	3.0	3.0	50.8	5.2	83.8		3.0	3.0	46.7	5.9	89.0
	3.0	4.0	49.3	5.3	83.8		3.0	4.0	50.3	4.4	89.0
	3.0	5.0	48.6	5.5	83.8		3.0	5.0	49.4	4.4	89.0
Grade III	1.0	1.0	67.9	4.4	48.8	Grade IV	1.0	1.0	75.1	3.5	54.5
	1.0	2.0	64.5	5.1	48.8		1.0	2.0	65.3	4.1	54.5
	1.0	3.0	56.1	6.0	48.8		1.0	3.0	64.7	4.2	54.5
	1.0	4.0	56.1	8.3	48.8		1.0	4.0	62.8	4.3	54.5
	1.0	5.0	55.4	8.7	48.8		1.0	5.0	60.9	4.6	54.5
	2.0	1.0	58.9	5.9	68.5		2.0	1.0	57.9	5.2	76.5
	2.0	2.0	50.1	6.4	68.5		2.0	2.0	57.8	3.1	76.5
	2.0	3.0	48.4	7.4	68.5		2.0	3.0	58.0	3.7	76.5
	2.0	4.0	46.8	5.8	68.5		2.0	4.0	54.9	4.9	76.5
	2.0	5.0	48.6	5.6	68.5		2.0	5.0	52.4	5.3	76.5
	3.0	1.0	51.0	9.0	89.0		3.0	1.0	55.2	5.1	99.5
	3.0	2.0	47.8	7.2	89.0		3.0	2.0	46.5	7.3	99.5
	3.0	3.0	45.3	6.1	89.0		3.0	3.0	50.6	3.9	99.5
	3.0	4.0	45.1	5.7	89.0		3.0	4.0	49.6	4.8	99.5
	3.0	5.0	45.7	4.7	89.0		3.0	5.0	49.0	4.4	99.5

cifically, subjective weights were assigned with values of 0.40, 0.40, 0.10, and 0.10 for $T_{\max}$, PCM weight, ΔT , and air velocity, respectively. To integrate the two weighting perspectives, a fusion coefficient (α) was introduced to control the contribution ratio between the objective and subjective components.

In this study, the subject weight proportion was set to 0.7 ($\alpha = 0.3$), indicating that subjective experience plays a dominant yet not exclusive role in determining the final weights. The fusion weights were calculated according to:

$$w_j = \alpha w_j^{(o)} + (1 - \alpha) w_j^{(s)} \quad (27)$$

where $w_j^{(o)}$ and $w_j^{(s)}$ represent the objective and subjective weights, respectively. The objective weights, subjective weights and final fusion weights are visualized. As shown in Fig. 9, $T_{\max}$ and PCM weight receive the highest fusion weights (both 0.35), confirming that thermal safety and lightweight design are the primary criteria for evaluating the BTMS. Air velocity has a moderate fusion weight (0.20), indicating that convective cooling and fan power consumption still play a noticeable role in the comprehensive assessment. In contrast, the temperature difference ΔT shows the smallest fusion weight (0.09). This suggests that, within the studied range, thermal non-uniformity is relatively less critical than the absolute peak temperature and system mass. These results demonstrate that the entropy-subjective fusion approach provides a balanced representation of data-driven performance sensitivity and engineering judgment, forming a reliable basis for the subsequent TOPSIS evaluation.

5.3.3. Comprehensive evaluation using TOPSIS method

Based on the fusion weights obtained from the entropy-subjective weighting process, the TOPSIS method was employed to conduct a comprehensive performance evaluation of the 60 cases. For each case, a normalized decision matrix was constructed using the selected performance indicators, including maximum temperature ($T_{\max}$), maximum temperature difference (ΔT), PCM weight, and air velocity. The relative closeness of each alternative to the ideal solution was calculated to obtain the TOPSIS score, which reflects the overall performance of each design scheme.

Fig. 10 presents the ranking results of the top 20 configurations according to their TOPSIS scores. The scores range from 0.16 to 0.66, indicating noticeable performance variations among the different parameter combinations. Schemes with higher scores (e.g., Rank 1–2) exhibit a balanced combination of low $T_{\max}$, relatively small ΔT , moderate system weight and low air velocity. This suggests that they achieve both effective cooling and good thermal uniformity without excessive mass and energy consumption. In contrast, the lower-ranked cases (Rank

>15) generally correspond to configurations with poor heat dissipation or excessively heavy PCM loading, which reduce the overall performance. The optimal configuration (Rank 1, TOPSIS score = 0.66) demonstrates superior comprehensive thermal management capability, characterized by a low peak temperature and small temperature gradient while maintaining relatively low system weight. These results provide valuable guidance for the multi-objective optimization of UAV battery thermal management systems, offering a practical balance between thermal safety, uniformity, and lightweight design.

Configurations and performance of top-ranked schemes were listed in Table 5. To further visualize and compare the performance characteristics of the top-ranked configurations, a radar chart was constructed (Fig. 11). The four axes represent the normalized values of maximum temperature ($T_{\max}$), temperature difference (ΔT), PCM weight, and air velocity. As shown in Table 5, all top five configurations maintain relatively low $T_{\max}$, indicating good thermal control within the battery module. The Rank 1 configuration (PCC Grade II, $s = 2$ mm, $v = 4$ m/s) shows a small polygon area, reflecting a balanced performance among all indicators. This combination achieves effective heat dissipation while minimizing PCM weight and air velocity, resulting in a lightweight and energy-efficient design. Comparatively, Rank 2 and Rank 3 configurations exhibit similar thermal performance but require higher PCM weight. As a result, the radar chart highlights that an appropriate combination of medium thermal conductivity PCC (Grade II), moderate spacing (2 mm), and optimized air velocity (4 m/s) yields the most desirable balance between thermal safety, uniformity, and lightweight design. As shown in Table 5, in scheme 1 (Rank 1), $T_{\max}$, ΔT , and weight of PCM are 46.8 °C, 5.8 °C, and 68.5 g, respectively. Compared with battery pack without cooling, the maximum temperature decreases by 43.3%, while the total system weight increases by 22.2%. These findings are consistent with the TOPSIS ranking results, further validating the reliability of the proposed multi-objective evaluation framework.

5.3.4. Sensitivity analysis of α -weighted model

The sensitivity analysis of the parameter α , which represents the relative importance assigned to entropy-based weights, is illustrated in Fig. 12. As α increases from 0 to 1, the weighting preference gradually shifts from subjective weights to purely entropy-driven weights. The TOPSIS scores of the top five schemes exhibit distinct trends as α varies. As $\alpha = 0$, subjective weights dominate the evaluation, and Scheme 1 achieves the highest TOPSIS score. This indicates superior performance under criteria that emphasize practical engineering considerations, such as lower maximum temperature and compact system design. However, as α increases, the influence of objective entropy weights becomes more prominent, resulting in a gradual decline in the TOPSIS score of Scheme 1. In contrast, Schemes 2 and 5 display mild upward trends in their scores with increasing α , suggesting improved performance under more objective evaluation criteria. Nonetheless, even as α approaches 0.8, Scheme 1 consistently remains among the top-performing configurations and is never displaced from the leading group. This demonstrates that Scheme 1 not only exhibits a clear advantage when subjective preferences dominate but also maintains strong competitiveness under objective weighting conditions. It is noteworthy that while objective (entropy-based) weights provide data-driven balance among indicators, the weighting scheme at $\alpha = 0.3$ aligns more closely with realistic engineering priorities. Within this parameter set, Scheme 1 continues to be identified as the optimal design, characterized by PCC Grade II, a 2 mm inter-cell spacing, and a 4 m/s airflow velocity. This finding further confirms the robustness and reliability of the proposed weight-fusion approach in balancing subjective and objective considerations within multi-criteria decision-making.

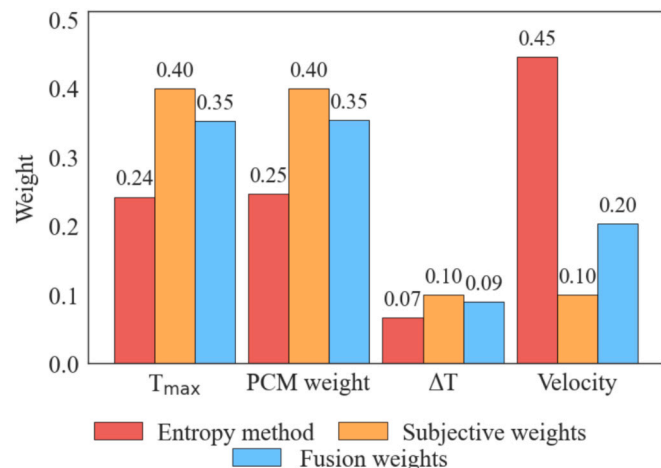


Fig. 9. Comparison of Weights for Various Indicators.

6. Conclusions

In this study, a lightweight and energy-efficient PCM–air hybrid BTMS for UAV applications was proposed and systematically optimized

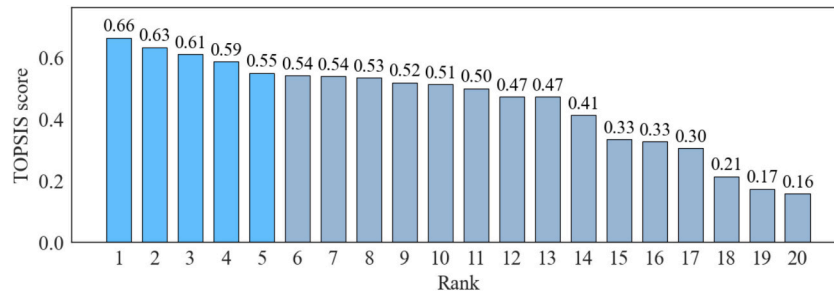


Fig. 10. Distribution of TOPSIS scores.

Table 5

Configurations and performance of top-ranked schemes.

Rank	Spacing (mm)	PCC Grade	Velocity (m/s)	T_{max} (°C)	ΔT (°C)	PCM weight (g)
1	2	Grade II	4	46.8	5.8	68.5
2	3	Grade II	3	45.3	6.1	89.0
3	3	Grade II	4	45.1	5.7	89.0
4	2	Grade I	4	47.9	5.3	64.4
5	2	Grade II	3	48.4	7.4	68.5

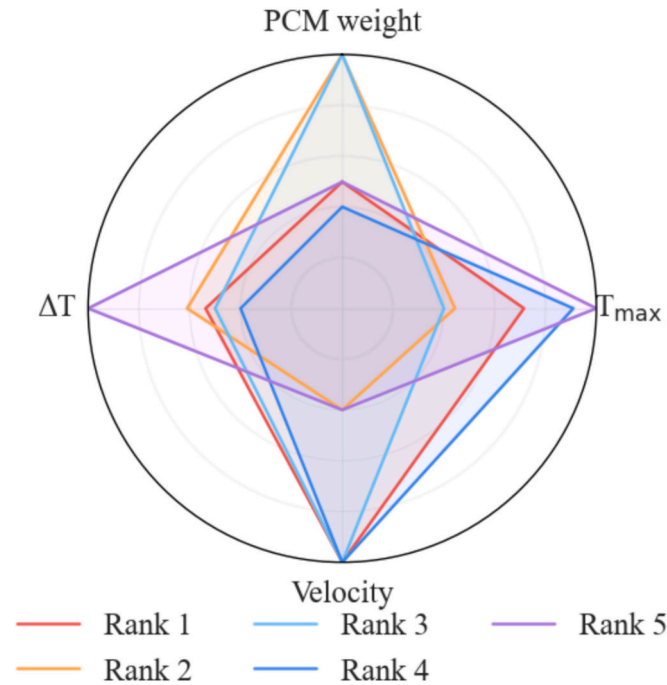
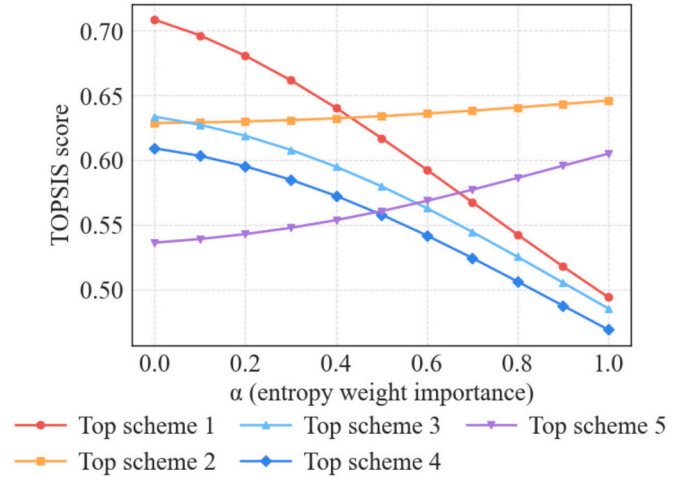


Fig. 11. Performance characteristics of the top-ranked schemes.

through coupled thermal modeling and multi-objective evaluation. The major conclusions were summarized as follows:

- (1) Single-factor analysis demonstrated that the thermal performance of the hybrid system is jointly governed by PCC properties, inter-cell spacing, and airflow velocity. Moderate enhancement of these parameters improves cooling effectiveness, whereas excessive increases provide limited additional benefit.
- (2) Multi-factor investigation revealed significant coupling effects among structural and operating parameters, highlighting the necessity of integrated optimization rather than single-parameter enhancement.
- (3) The entropy-subjective weighted TOPSIS framework successfully identified an optimal configuration that achieves substantial

Fig. 12. Sensitivity analysis of α for TOPSIS stability.

temperature reduction while maintaining a reasonable increase in system mass, demonstrating an effective balance between thermal safety and lightweight requirements.

- (4) Sensitivity analysis of the weighting strategy confirmed the robustness and stability of the optimization results under varying subjective-objective preference scenarios.

Overall, this study establishes an engineering-oriented multi-objective decision-making framework for lightweight UAV battery thermal management. Rather than focusing solely on material innovation, the proposed approach emphasizes system-level trade-offs among thermal regulation efficiency, structural compactness, and mass penalty, providing practical guidance for UAV-oriented BTMS design. More importantly, the integrated multi-objective optimization framework developed in this study offers a transferable methodology for the design and evaluation of next-generation lightweight BTMS for compact aerial platforms.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.applthermaleng.2026.130845>.

Data availability

Data will be made available on request.

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